Conformal Corrective RAG

Why and What | Mini Project Idea | Prompt Engineering and Optimization for Generative AI (UE24AM343AB5)

Why

Corrective RAG (CRAG) improves on standard retrieval-augmented generation by grading retrieved documents and routing to generation, query rewriting, or web search depending on relevance. But CRAG's grading thresholds are hand-set by trial and error, so there is no guarantee on how often the system is wrong. More importantly, CRAG never says "I don't know": even when retrieval and the web fallback both return poor context, it still generates an answer, and recent work (Sufficient Context, ICLR 2025) shows this is exactly when models hallucinate instead of abstaining.

At the same time, conformal prediction methods can turn a model's raw confidence score into a calibrated decision with a statistical guarantee on the error rate, but so far this has only been applied to detecting hallucinations after the fact, not to deciding which corrective action a RAG pipeline should take. There is a real gap between "routing CRAG" and "calibrated, abstention-aware RAG," and no surveyed work currently bridges it.

What

We extend CRAG with a calibration layer and a fourth action. A sufficiency grader scores each retrieved set; a conformal calibrator (fit on a held-out set) converts that score into a threshold that keeps the error rate of answered queries below a chosen risk level, instead of a guessed cutoff. Based on the calibrated score, the system routes to one of four actions:

- **Generate** – score is high, retrieved context is used directly.
- **Rewrite and re-retrieve** – score is mid-range, the query is reformulated and retrieval repeated.
- **Web fallback** – score is still low after repair (open-domain mode only; disabled in closed-domain mode to respect data-governance limits).
- **Abstain** – score remains low even after rewriting and web search, so the system declines to answer rather than guess.

We evaluate this pipeline under a controlled noise-injection benchmark (irrelevant, outdated, and partial retrieval) against standard RAG, vanilla CRAG, and sufficiency-only abstention baselines, measuring accuracy, hallucination rate, abstention rate, coverage at a fixed risk level, latency, and web calls. The contribution is twofold: a calibrated, abstention-capable corrective router, and a reusable noise-injection benchmark for stress-testing RAG pipelines under retrieval failure.